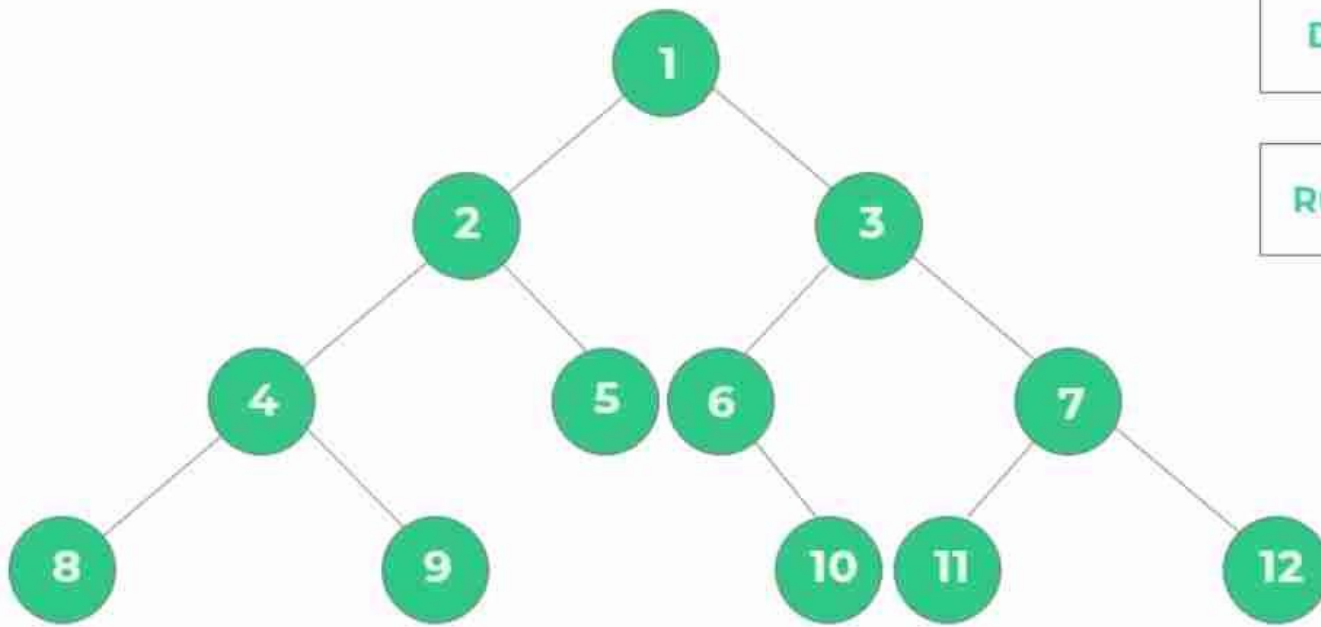


Tree

Question No:1

DS	Session	Tree 1	Question Information	Level 1	Challenge 61
Question description Jegan Sir students were chatting and playing quite loudly on the last day of the year, celebrating the end of the academic session. Jegan sir was harshly chastised by the college's principal. But, instead of becoming enraged, he attempted to engage everyone in a different task. So Jegan sir gave his students to solve the task such that, you have to perform pre-order tree traversal in Binary search tree.					



Direction : Anti Clock

Rule : CLR (Center-Left-Right)

PreOrder:

1 2 4 8 9 5 3 6 10 7 11 12

Constraints:

0<size <100

0<data<1000

Input format:

First line indicates the size of the tree

Second line indicates the elements of the tree

Output Format:

single line indicates the pre-order list of the given value

Logical Test Cases

Test Case 1

INPUT (STDIN)

7
200 120 80 160 280 240 320

EXPECTED OUTPUT

200 120 80 160 280 240 320

Test Case 2

INPUT (STDIN)

9
34 67 10 90 410 810 40 20 60

EXPECTED OUTPUT

34 10 20 67 40 60 90 410 810

Mandatory Test Cases

Test Case 1

KEYWORD

void preorder(struct node* root)

Test Case 4

KEYWORD

struct node *left,*right;

Test Case 2

KEYWORD

struct node* newNode(int item)

Test Case 3

KEYWORD

struct node

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

280

Test Case 3

NLOC

50

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  struct Node {
4      int data;
5      struct Node *left;
6      struct Node *right;
7  };
8  // Create a new node
9  struct Node* createNode(int data) {
10     struct Node* newNode =
11         (struct Node*)malloc(sizeof(struct Node));
12
13     newNode->data = data;
14     newNode->left = NULL;
15     newNode->right = NULL;
16
17     return newNode;
18 }
19 // Insert into BST
20 struct Node* insert(struct Node* root, int data) {
21
22     if (root == NULL) {
23         return createNode(data);
24     }
25
26     if (data < root->data) {
27         root->left = insert(root->left, data);
28     }
```

```
29         else if (data > root->data) {
30             root->right = insert(root->right, data);
31         }
32
33         return root;
34     }
35     // Preorder: Center -> Left -> Right
36     void preorder(struct Node* root) {
37
38         if (root == NULL) {
39             return;
40         }
41
42         printf("%d ", root->data);
43
44         preorder(root->left);
45
46         preorder(root->right);
47     }
48     int main() {
49
50         int n;
51         scanf("%d", &n);
```

```

53     struct Node* root = NULL;
54
55     for (int i = 0; i < n; i++) {
56         int value;
57         scanf("%d", &value);
58
59         root = insert(root, value);
60     }
61
62     preorder(root);
63     return 0;
64 }

```

Outout:

Test Case-1:

```

7
200 120 80 160 280 240 320
200 120 80 160 280 240 320

```

Test Case-2:

	9	
	34 67 10 90 410 810 40 20 60	
	34 10 20 67 40 60 90 410 810	

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